

Family 10: one schema for every granularity

a storage and sampling design in which no source is resampled to a common grid

blauewelt/earth — design note

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Abstract

The programme’s input tensors so far share one contract: every source is brought onto a 0.25° , five-day grid before the model sees it. That contract copies coarse values into fine cells as if they were independent measurements, averages fine sources down to the grid and discards what they were bought for, and represents a sparse observing system as a grid that is mostly empty. Family 10 replaces it with three commitments. Every source is stored at its own resolution and cadence in one of three tiers by density: gridded dense fields at their native grid, points and profiles in a pentad-sorted observation store, and fine imagery as a catalogue plus the cached tokens of a small per-tile codec. A single token schema — the value, its offset from the anchor in space and time, and its *footprint* (the spatial and temporal support it averaged, in log units) — is produced by each tier’s reader, so the encoder never learns which file a value came from, only what it is, where it is, and how much it averaged. And the decoder is queried at a location and a footprint, so targets can be at any granularity too. Families 7.1 and 8 become the first tier-G and tier-P groups of family 10 unchanged. We state the schema, the tiers, the registry that binds them, the local codec, the storage arithmetic, and four staged experiments each with a falsifier.

1 Why a common grid is the wrong contract

Three properties of the existing tensors motivate the design; each is a measurement rather than a preference.

A coarse value copied into fine cells is one measurement presented as many. Family 7 serves a reanalysis value on a 1.9° Gaussian cell, stored at 1° , to the sixteen 0.25° cells inside it and to the nine cells of a lag-0 patch, as sixteen or nine tokens that happen to agree. The model cannot distinguish nine readings of 0.31 from one reading copied nine times, and those are different amounts of evidence.

A fine value averaged into a coarse cell discards what the finer source adds. Family 7.1 keeps one thirty-sixth of the pixels of a 4 km colour product and adds a coverage channel to say how much was discarded. Kilometre-scale radiometry is of order 3.5×10^{14} values; it cannot be a tensor, and “therefore average it” is the wrong conclusion.

Distance is a feature that a grid hides. With the same five nearest Argo profiles, optimal interpolation beats climatology by 21 % on temperature where the gridded monthly product, read at the nearest cell, gave a learned codec 3 % (E-076b). The information was in where and when the profiles were taken relative to the point being predicted; the grid had removed it.

The contract cannot therefore be “everything on one grid”. It has to be “everything at its own granularity, and the token says what that granularity is”.

2 The token

Every observation, from every tier, becomes one token:

$$\text{token} = (\mathbf{v}[C], \mathbf{m}[C], \Delta x, \Delta y, \Delta t, \log_2 f_p, \log_2 f_t, n_R, s, c) \quad (1)$$

where $\mathbf{v}$ is the measurement, standardised per channel, with mask $\mathbf{m}$ marking which of its C components were measured; $(\Delta x, \Delta y)$ the offset from the anchor in kilometres east and north and Δt the age in days, measured back from the end of the anchor’s bin; n_R the number of such observations within the search bounds; s and c learned embeddings of the source class and the channel. The two footprint fields are

$$\log_2 f_p = \log_2 \frac{\text{footprint in km}}{27.83 \text{ km}}, \quad \log_2 f_t = \log_2 \frac{\text{support in days}}{5 \text{ days}}, \quad (2)$$

the spatial and temporal support in units of a 0.25° cell and a pentad, clamped to ± 12 . Table 1 places the programme’s sources on that scale. For a dense group both fields are constants and cost nothing; for a sparse store they are per row.

source	$\log_2 f_p$	$\log_2 f_t$
Sentinel-2 pixel, 10 m, instantaneous	−11.4	−4 (clamped)
Argo profile (point, minutes)	−4 (clamped)	−4 (clamped)
OLCI pixel, 300 m, daily	−6.5	−2.3
OC-CCI cell, 4 km, daily	−2.8	−2.3
OISST cell, 0.25° , pentad mean	0	0
a 1° box average, pentad	+2	0
NCEP/NCAR at its T62 support ($\approx 1.9^\circ$), pentad	+2.9	0
Roemmich–Gilson Argo field, 1° , monthly	+2	+2.6

Table 1: The footprint fields for the programme’s sources. The reanalysis is entered at the resolution it resolved, not the grid it is stored on.

The offsets are the correction to the first property above: a 1° value is one token located at the cell’s centre, seen from sixteen different offsets by the sixteen anchors inside it, and a fine anchor’s lag-0 patch holds one such token rather than nine. The footprint fields are what the offsets cannot carry: a value 30 km away from a 1.9° average and one 30 km away from a point profile say different things about the anchor, because the coarse value is a smoother, lower-variance quantity. This is the choice AlphaEarth makes on the encoder side, conditioning each source’s decoder on the sensor’s own metadata rather than resampling to a common grid first [1].

3 Three tiers, one registry, one sampler contract

Sources are stored by density, because density decides what a read costs.

tier	what	layout	reader
G — gridded dense ($\gtrsim 4$ km, daily to monthly, near-complete)	GLORYS, OISST, NCEP/ERA5, OC-CCI, DUACS, GREP levels	one bin-major array per group at its native grid, own first bin or bin list, standardised, NaN = unobserved	a slab read plus a block or nearest lookup at the anchor; footprint a per-group constant. Exists: families 7 and 7.1
P — points, profiles, tracks (sparse in space or time)	Argo core, Deep and BGC, drifters, moorings, tide gauges, altimeter tracks, SWOT swath samples, lidar footprints, SOCAT	columns sorted by pentad bin with CSR offsets; raw units; per row (lat, lon, t , footprint, $\mathbf{v}[C]$, $\mathbf{m}[C]$)	k nearest in a space–time metric, one-sided in time, bounded by $(R_{\max}, T_{\max})$, miss tokens to a fixed k . Exists: family 8
T — tiles (fine imagery, $\lesssim 1$ km, swaths)	Sentinel-1/2, Landsat, NISAR, OLCI/SLSTR, VIIRS, MODIS 1 km	not stored as values: a tile-date catalogue (footprint polygon, sensor, time, valid fraction, URL) plus a token cache — the local codec’s output per tile-date — stored as a tier-P store whose values are the tokens	the tier-P reader; a cached token is an observation with the tile’s footprint and $\log_2 f_t = -4$. Not yet built

Table 2: The three storage tiers.

Three properties make the tiers one family. *One time axis*: every tier is indexed by the pentad bin from the 1982-01-01 epoch; tier-P rows keep their timestamp inside the bin so Δt is fractional; a tier-G group carries its own first bin or, for a monthly product, its own list of live bins. *One registry*: a `family.json` at the family’s root lists every group with its tier, its grid or store schema, its cadence,

its channel names and units, its footprint constants, its first bin, its file names with hashes, its sources and the builder commit; consumers dispatch on the tier, so a new group is a new entry rather than a new code path. *One sampler contract:* `gather(anchor,family) → tokens` — for each group, its reader returns tokens in the schema of §2 under that group’s cone family (the reach-versus-lag rule of E-071: atmosphere, currents and sea-surface height, temperature and mixed layer, land, the depth column) with a fixed slot count per group, so batches stay rectangular. The existing cone sampler is this contract for tier G, and the Argo store’s nearest-neighbour search is it for tier P; tier T reuses tier P’s.

4 The local codec

A 64×64 -pixel tile at 10 m is 4,096 pixels of 13 bands; a 0.25° cell holds about 7,700 such tiles. They cannot be dots. The local codec is the programme’s per-pixel masked-autoencoder codec generalised to a tile: input one tile-date — all bands, with the sensor’s own metadata (resolution, incidence angle for radar, sun angle for optical) as conditioning — and output $k = 4\text{--}8$ tokens of 64 dimensions, trained by masked-pixel reconstruction plus a same-place-different-date term so that the token carries state rather than texture. Tokens are cached once per tile-date and reused by every anchor whose cone contains the tile. The codec has snapshot semantics; the cone codec supplies the time axis by reading the cached tokens of many dates. The hierarchy is: raw observations $\rightarrow$ local codec (per tile-date, cached) $\rightarrow$ cone codec (per anchor, over local tokens, coarse dots and point observations) $\rightarrow$ task heads. AlphaEarth’s space–time pyramid does this within one image [1]; here the pyramid runs across sources and time scales, and the set encoder is Perceiver-style cross-attention from a fixed number of latents to the token set [2], so cost is linear in the number of tokens and a ragged input — a profile here, a swath there, a tile in the corner — needs no padding.

Storage arithmetic. Over land and coasts ($\approx 1.5 \times 10^8 \text{ km}^2$) there are $\approx 3.7 \times 10^8$ tiles of 640 m; at ≈ 70 clear dates per year, 6 tokens of 64 float16 dimensions, that is $\approx 20 \text{ TB}$ of tokens per year — streamable as a tier-P store but not small. Tiles of 256 pixels (2.56 km) cut it sixteenfold to $\approx 1.3 \text{ TB}$ per year and remain a hundred times finer in area than a 0.25° cell; the tile size is a parameter the first experiment measures. The catalogue is metadata only, of order 10 GB per year. Raw imagery is streamed once to train the codec and once to encode, and is never kept.

Radar. NISAR’s L-band backscatter at 15 m on a 12-day repeat, through cloud and at night, is the right snapshot input for soil moisture, flooding, ice and vegetation structure, and complements Sentinel-1’s C band; its record begins in late 2025, so it enters as a local-codec input and cannot contribute to a 43-year forecast axis. Sentinel-1 (2014–) and Sentinel-2 (2015–) carry the record. One codec conditioned on sensor metadata serves both radars with different source embeddings, which is the reason the codec is conditioned rather than trained per sensor.

5 Outputs at any granularity

The target side has the same problem as the input side. The recipe already has a decoder queried at a location (E-072); it gains the two footprint fields of the token. “The mean chlorophyll of this 0.25° cell over this pentad” and “this 300 m pixel on this day” are then two queries of one embedding, each scored against the observation that actually has that support. A masked-token objective over the schema of §2 covers every tier at once: withhold a token — a grid cell, a profile, a cached tile token — and predict its value from the rest, at its own footprint. Family 8’s two objectives, masked-profile reconstruction and next-pentad prediction, are the tier-P instances of it. Every read-out remains a head on the frozen representation plus the rolled forecast, and is scored against the null ladder — climatology, damped persistence, a linear inverse model in pixel and in embedding space — the programme’s protocol requires.

6 What is built first

Four steps, each with its falsifier stated before it is run.

1. **The footprint fields on the tensor that exists.** Read family 7’s coarse groups as offset tokens — one token at the coarse cell’s centre, its $(\Delta x, \Delta y)$ to the anchor, $\log_2 f_p = +2.9$ for the reanalysis and +2 for the monthly Argo field — instead of nearest-cell copies, and add the footprint fields to every sparse token. One cone-codec arm against its twin on the same anchors and seed. Falsifier: no change in held-out per-family loss, in which case the model does not use the fields at this scale and the tile tier must be argued on its own evidence. Cost: a sampler change and about one dollar of compute.
2. **The registry and the tier dispatch.** Write `family.json` for families 7.1 and 8 as one family, make the samplers and the front end dispatch on the tier, retire the per-family metadata conventions. No model. Falsifier: a test that gathers one anchor through the registry and reproduces the existing sampler’s token bytes exactly.
3. **The first local codec, coastal.** Sentinel-2 level-2A 10 m tiles over about three hundred coastal 0.25° cells where colour, sea-surface temperature and Argo all have coverage; train the tile codec, cache its tokens as a tier-P store, add them to the cone; three seeds, since a new tier buys its own replication. Verdict: the masked-token loss on the colour and temperature targets at those anchors with and without the tile tokens. Falsifier: parity, in which case at pentad cadence the coarse fields already carry what 10 m adds and the fine tier is for fine targets, not for the forecast. This is the one compute-bound step.
4. **NISAR and Sentinel-1 through the same codec** with a radar source embedding, over sea-ice-edge and flood-prone cells where the day-and-night, through-cloud property matters. After step 3.

7 What does not change

The pentad bin as the unit of time; the holdout protocol (2009, 2017 and 2023 as development years; train through 2020, test 2021–2024); the cone families and reaches; family 7.1’s files, which become tier-G groups of family 10 unchanged; family 8’s store, which becomes its first tier-P group unchanged. The design adds a registry and a tier and removes nothing.

References

- [1] Brown, C. F. et al. (2025). AlphaEarth Foundations: An embedding field model for accurate and efficient global mapping from sparse label data. arXiv:2507.22291.
- [2] Jaegle, A. et al. (2021). Perceiver IO: A General Architecture for Structured Inputs & Outputs. arXiv:2107.14795.
- [3] blauewelt/earth (2026). Family 7: a global, three-resolution input tensor. Design note, 7 September 2026.
- [4] blauewelt/earth (2026). Family 7.1: ocean colour joins the global tensor. Design note, 13 September 2026.
- [5] blauewelt/earth (2026). E-076: family 8 — the measurement is never missing, only far away. Plan and results, 7–8 September 2026.